

DO AI WEATHER MODELS LOSE SKILL IN THE TAILS? A STATION-BASED COMPARISON OF GLOBAL AND REGIONAL SYSTEMS

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ABSTRACT

AI weather models are often reported to underpredict extremes, but most evidence comes from deterministic global systems verified against reanalysis. We compare eleven learned and numerical alternatives to ECMWF IFS over ten months. Wind and temperature use 667 European stations at 6–48 h leads; solar radiation uses a separate European station panel and hourly leads. Regimes are fixed by an ERA5 1991–2020 climatology. After a rolling domain-mean bias correction applied identically to every system, all five global learned systems lose 2.7–6.8 points of relative MAE skill from all conditions to the >P95 regime in both variables, as does the ECMWF ENS mean. Under these fixed climatological regimes, two regional generative systems and the deterministic global numerical models do not. Without correction, both regional systems still gain and four of five global learned systems still lose temperature tail skill, but several wind rankings change. Bias conditioned on the observation points toward the centre for every system, as expected for imperfect forecasts. When conditioning on the forecast instead, high-tail temperature bias is within 0.3 °C of zero for every system, while wind reveals larger model-specific differences. The station-verified tail deficit is therefore reproduced for the tested global learned cohort under fixed climatological regimes, weakens under in-window percentiles, and is not uniform across learned forecasting systems. The solar-specific Helios model is strongest in both irradiance tails but loses skill in the centre of the distribution, showing that specialization redistributes rather than uniformly improves skill.

1 INTRODUCTION

Data-driven weather models now match or outperform operational numerical systems on standard forecast scores (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). Their behaviour in the tails remains contested. Models trained with mean squared error are expected to regress toward the conditional mean and smooth sharp gradients (Olivetti & Messori, 2024a; Bonavita, 2024). Storm Ciarán forecasts underestimated peak winds (Charlton-Perez et al., 2024), and errors for record-breaking heat, cold, and wind are systematically larger than IFS HRES (Zhang et al., 2026). Other events and regions show learned systems matching or exceeding numerical forecasts (Olivetti & Messori, 2024b; Pasche et al., 2025).

Three design choices limit what can be inferred from this literature. First, many studies verify ERA5-trained models against ERA5, whereas station observations expose local variability absent from a gridded target (Jin et al., 2024; Gupta et al., 2025). Second, most comparisons concern global deterministic regression systems. Regional learned models and proper-score-trained ensembles have been studied individually (Nipen et al., 2026; Lang et al., 2026), but rarely under the same station-based protocol. Third, evaluation only on cases where the observation is extreme creates the forecaster’s dilemma: it rewards high-amplitude forecasts and induces apparent regression toward the centre for any imperfect forecast (Murphy & Winkler, 1987; Lerch et al., 2017).

We therefore ask a bounded question: *after the same rolling mean-bias correction is applied to every forecast, do learned systems lose relative MAE skill from all conditions to the tails, and does that loss separate learned from numerical forecasting?* We compare global and regional systems

over Europe for 10 m wind and 2 m temperature. Observed-value regimes use fixed ERA5 1991–2020 thresholds at each station, day of year, and hour. We report absolute skill relative to IFS, the change from all-condition to tail skill, raw-versus-corrected sensitivity, and both observation- and forecast-conditioned bias.

The answer is structured rather than uniform. Under fixed climatological regimes and after correction, every tested global learned system loses relative skill from all conditions to the upper tail in both variables. The ECMWF ENS mean behaves similarly. The two regional generative systems and deterministic global numerical models do not, although the regional wind result is sensitive to bias correction and regime definition. This pattern does not isolate a cause: domain, resolution, training data, ensemble averaging, and model provenance are confounded. It does identify which systems reproduce the reported tail deficit and which provide counterexamples.

2 RELATED WORK

Evidence at extremes. Storm Ciarán evaluations found that FourCastNet, Pangu-Weather, and GraphCast reproduced the synoptic structure but underestimated peak winds (Charlton-Perez et al., 2024). Record-breaking heat, cold, and wind were also forecast less accurately by GraphCast, Pangu-Weather, and FuXi than by IFS HRES (Zhang et al., 2026). Systematic results are more mixed. Against ERA5, GraphCast and Pangu-Weather compete with HRES in many regions but lose for warm extremes over Europe (Olivetti & Messori, 2024b). At European SYNOP stations, Pangu-Weather discriminates summer warm events well, while still missing some intensity extremes (Ben Bouallègue et al., 2024). Impact case studies likewise find learned forecasts competitive locally on one heatwave and superior on a compound winter storm, but weaker under other aggregations (Pasche et al., 2025).

Objectives and regional systems. Grid-point regression incurs a double penalty when a displaced feature is penalized both where it is missing and where it appears. This can reduce the effective resolution of learned forecasts (Subich et al., 2025). Generative models instead learn a distribution of plausible states (Price et al., 2025; Pathak et al., 2026). AIFS ENS optimizes an objective based on a proper score (Lang et al., 2026), while regional systems increase spatial resolution over a limited domain (Nipen et al., 2026; Molinaro et al., 2026). In this study every ensemble is evaluated through its mean. The comparison therefore tests the tail error of a point summary, not whether a probabilistic objective recovers the distribution of extremes.

Station verification. WeatherReal (Jin et al., 2024) and MAUSAM (Gupta et al., 2025) show that near-surface errors against observations can substantially exceed errors against reanalysis. We use the StationBench stack (Schlueter & Wagner, 2025), but define regimes from an external climatology rather than from the scored period.

Conditional verification. Forecast verification admits two complementary factorizations: observations conditioned on forecasts diagnose reliability, while forecasts conditioned on observations describe discrimination (Murphy & Winkler, 1987). Restricting a conventional score to extreme observations is not a proper forecast comparison and can favour exaggerated amplitude (Lerch et al., 2017). We retain observed regimes because they directly measure error on the realized tail, but interpret them comparatively and add forecast-conditioned bias as a reliability diagnostic.

3 METHODS

3.1 FORECAST SYSTEMS AND OBSERVATIONS

Track A compares ten wind and temperature alternatives with ECMWF IFS: ECMWF ENS, NOAA GFS, DWD ICON Global, ECMWF AIFS (Lang et al., 2024), ECMWF AIFS ENS (Lang et al., 2026), Microsoft Aurora (Bodnar et al., 2025), and Jua EPT-2 Reasoning, EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa (Molinaro et al., 2025; Siegenheim, 2025; Jua AI, 2026). AIFS ENS is scored as a 51-member mean; EPT-2 HRRR and EPT-2.1 Europa are diffusion-based ensembles. The solar comparison additionally includes the solar-specific EPT-2.1 Helios (Molinaro, 2026). Track B adds

the regional numerical ICON-EU on the March–June 2026 window. Appendix A lists model class, output, domain, and archive provenance.

Forecasts are verified at 667 stations with complete climatological thresholds across thirteen European countries. The scored panel contains 227 stations in France and 8–81 in each other country (Table 2). Gridded forecasts are bilinearly interpolated to station locations. The EPT-2 family was pretrained partly on SYNOP and METAR networks (Jua.ai, 2026), so station-network independence is not claimed; the forecast-valid period follows the published pretraining period. Model results are out of sample with respect to model development, but station overlap remains a limitation.

Track A covers 1 September 2025–30 June 2026. Wind and temperature use all 00, 06, 12, and 18 UTC initializations and the common 6, 12, ..., 48 h lead grid. A model and IFS are pooled on their common country, regime, lead, and initialization-time cells, using the same station–initialization–lead pairs within each cell. Results are sample-weighted across countries, so France contributes 39% of upper-wind-tail and 48% of upper-temperature-tail samples.

Solar radiation is evaluated on a separate quality-controlled panel over the same period and on hourly leads from 1 to 48 h. Regime-stratified results use 543 stations with fixed thresholds across nine countries. The model set contains Helios, Europa, HRRR, EPT-2e, EPT-2 Reasoning, ICON Global, GFS, and IFS. All reported solar results are likewise out of sample.

3.2 CLIMATOLOGICAL REGIMES

For each station, calendar day, and UTC hour, we estimate ERA5 P5, P25, P75, and P95 over 1991–2020 using a cyclic ± 15 -day window. Each threshold is estimated from 930 values. Forecast–observation pairs are assigned to a regime using the observation. The bands are $<P5$, $P5$ – $P25$, $P25$ – $P75$, $P75$ – $P95$, and $>P95$, together with all conditions.

These are anomaly regimes, not impact thresholds. Across station–time cells, the median wind P95 is 6.6 m s^{-1} ; median temperature P95 and P5 are 16.0°C and 5.6°C . Wind tails are approximately symmetric (10.2% above P95 and 9.2% below P5), consistent with greater point-observation variance. Temperature is asymmetric (13.5% above P95 and 3.2% below P5), reflecting a warm evaluation period relative to 1991–2020. We therefore use percentile notation rather than physical labels.

3.3 SKILL, DEBIASING, AND UNCERTAINTY

For model m in evaluation cell c , MAE skill relative to IFS is

$$S_{m,c} = 100 \left(1 - \frac{\text{MAE}_{m,c}}{\text{MAE}_{\text{IFS},c}} \right). \quad (1)$$

Positive values indicate lower MAE than IFS. Ensemble systems are reduced to their means; MAE of the mean does not assess ensemble spread or calibration.

The primary comparison removes a rolling domain-mean offset. For each model, week, initialization hour, and lead, we average errors over all European stations during the preceding four valid weeks and subtract that scalar from the next week’s forecasts. The correction does not use the observed regime, but a uniform shift can have a large regime-dependent effect. Raw and corrected results are therefore reported together. The correction is causal with respect to valid time; up to one forecast lead of its training window can postdate forecast issuance.

We pool MAE linearly by sample count and discard country–lead cells with fewer than 100 pairs. Uncertainty is one leave-one-month-out jackknife standard error, computed with the same linear pooling as the point estimate. Track A has ten monthly blocks and Track B four. These errors describe month-to-month sensitivity of the pooled estimate, not country-level dispersion or a multiple-testing-adjusted confidence interval.

3.4 ROBUSTNESS AND CONDITIONAL DIAGNOSTICS

Our direct test of tail degradation is $\Delta S = S_{>P95} - S_{\text{all}}$. Negative values mean that a system loses skill relative to IFS when moving from all conditions to the observed upper tail. We report ΔS for

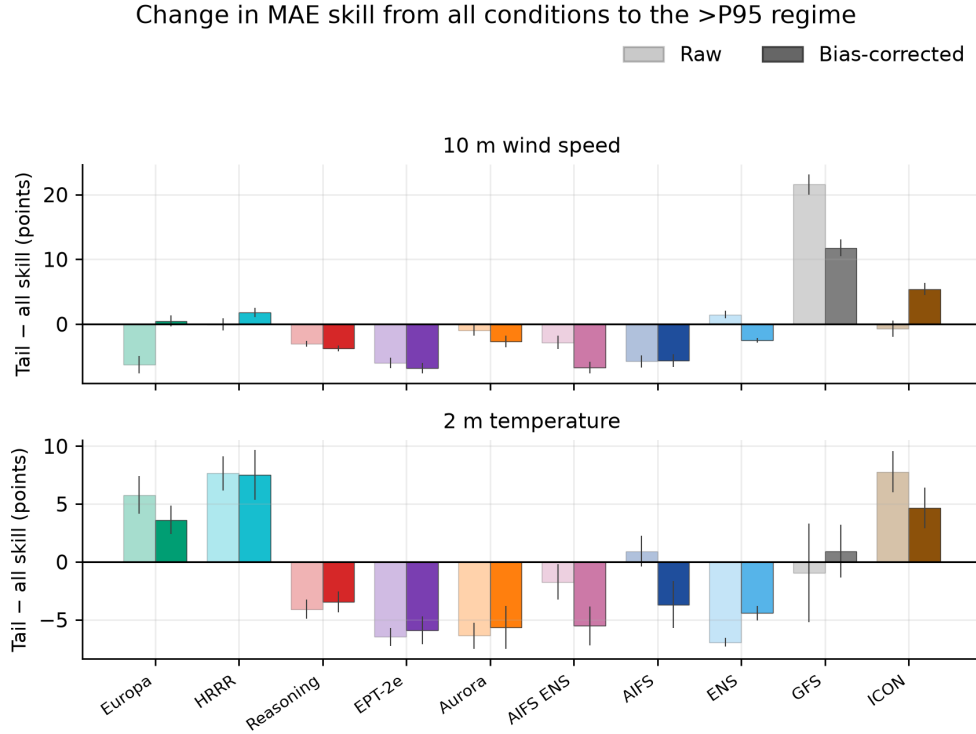


Figure 1: Change in MAE skill from all conditions to the observed >P95 regime, 6–48 h. Negative values indicate relative degradation in the tail. Light bars are raw forecasts; dark bars apply the common rolling domain-mean bias correction. Error bars give one leave-one-month-out jackknife standard error for the tail-minus-all difference.

raw and corrected forecasts. Appendix C additionally replaces the station–day–hour climatological regimes with one country-level percentile distribution over the evaluation window. This selects absolute rather than local-anomaly tails.

Observation-conditioned bias, $E[F - Y \mid Y \in \mathcal{R}]$, is expected to point toward the centre for an imperfect forecast. We therefore also report forecast-conditioned bias, $E[F - Y \mid F \in \mathcal{R}]$, using the same fixed thresholds. The latter is zero for a conditionally unbiased point forecast and provides the reliability view missing from an observed-tail score.

4 RESULTS

4.1 A STRUCTURED TAIL PENALTY

Figure 1 reports the change from all-condition to upper-tail skill. After correction, every global learned system loses 2.7–6.8 points in both variables. The ECMWF ENS mean loses 2.5 points in wind and 4.4 in temperature. HRRR gains 1.8 ± 0.7 points in wind and 7.5 ± 2.1 in temperature; Europa’s changes are $+0.5 \pm 0.8$ and $+3.6 \pm 1.2$. Deterministic ICON Global and GFS gain 0.9–11.8 points. Learned versus numerical forecasting does not separate the outcomes, but global learned systems and ensemble means form a consistent negative group.

The correction matters. On raw wind forecasts, Europa loses 6.2 points from all conditions to >P95, HRRR is approximately unchanged, and GFS gains 22 points. For temperature, both regional learned systems retain positive tail changes before and after correction, while raw AIFS gains 0.9 points in the upper tail. The regional exception is therefore more robust to correction in temperature than wind, whereas the global learned penalty is more robust in wind. No tail ranking is reported independently of the correction.

4.2 ABSOLUTE SKILL AND AIFS ENS

Absolute corrected skill remains heterogeneous (Figure 2). Europa and HRRR are ahead of IFS in the upper wind and temperature tails; ICON Global is also ahead. All global learned systems are behind IFS in upper-tail wind. In temperature, EPT-2 Reasoning remains slightly positive while AIFS, AIFS ENS, Aurora, and EPT-2e are negative. Below temperature P5, those four systems lose 20–24% under the climatological definition, although this magnitude is not robust to the threshold definition.

AIFS ENS narrows the all-condition wind deficit of deterministic AIFS (-2.4% versus -5.7%) but remains behind IFS above the wind P95 (-9.1%). Against the like-for-like ECMWF ENS mean, it is 7.9 points worse in that regime. For temperature, AIFS ENS and the ECMWF ENS mean have similar upper-tail skill (-5.1% and -6.1%). This result constrains the ensemble mean only: it does not test whether proper-score training improves tail probabilities.

4.3 SOLAR SPECIALIZATION

Solar results reinforce the need to compare regimes rather than one mean score. Helios leads all conditions ($+10.1 \pm 1.7\%$), P5–P25 irradiance ($+16.4 \pm 2.7\%$), and the $>P95$ regime ($+24.8 \pm 5.7\%$), but loses $8.2 \pm 1.2\%$ in the central P25–P75 band. EPT-2 Reasoning instead leads that central band at $+14.0 \pm 0.6\%$. GFS shows the same redistribution in more extreme form: $+16.7\%$ above P95 and -28.3% in the centre. Observation-conditioned tail skill therefore reflects amplitude as well as accuracy, consistent with the forecaster’s dilemma.

4.4 SENSITIVITY TO THRESHOLDS AND HORIZON

Changing from fixed climatological thresholds to in-window percentiles reduces several large tail magnitudes. In particular, AIFS ENS skill below temperature P5 changes from -21.3% to -0.3% , while its upper-tail wind and temperature results remain negative. The regional systems remain positive in both upper tails, but EPT-2 Reasoning becomes positive under the in-window definition. Under that definition, the wind penalties of AIFS and Aurora shrink to within one standard error of zero (-0.3 and -0.2 points), the temperature changes of Aurora and EPT-2 Reasoning turn positive, and the regional wind gain reverses to -2 to -3 points. The uniform sign is a property of the fixed climatological regimes. The claim is therefore about a structured tendency, not an invariant ordering for every definition (Table 11).

4.5 WHAT CONDITIONAL BIAS SHOWS

The left column of Figure 3 reproduces the displacement expected when conditioning an imperfect forecast on the observation. Every system overpredicts low observations and underpredicts high observations. The common sign is not evidence that models themselves shrink extremes.

The forecast-conditioned view is different. Above the temperature P95, every system lies within $0.3\text{ }^{\circ}\text{C}$ of zero bias. Above the wind P95, most systems lie within 0.5 m s^{-1} , while GFS overpredicts by 1.0 m s^{-1} . Forecasts in the lowest wind regime are too low by $0.4\text{--}1.1\text{ m s}^{-1}$. Thus the observation-conditioned fingerprint mainly reflects the verification factorization. Forecast-conditioned errors are smaller; some are shared, such as underprediction in the lowest wind regime, while upper-tail wind errors differ by model.

4.6 REGIONAL CORROBORATION

On Track B, EPT-2.1 Europa has $+6.5 \pm 0.3\%$ all-condition wind skill, compared with $+4.6 \pm 0.4\%$ for ICON-EU. Their upper-wind-tail estimates, $+9.3 \pm 1.5\%$ and $+10.4 \pm 1.8\%$, are close relative to a four-month jackknife. The learned and numerical regional systems are also positive in the upper temperature tail. This comparison supports a regional exception but does not isolate resolution, model design, or provenance.

5 DISCUSSION

The evidence supports a narrower conclusion than a binary claim about AI weather models. After domain-mean bias removal, every tested global learned system loses relative MAE skill from all conditions to the upper wind and temperature tails under the fixed climatological regimes. Two regional learned systems do not, and regional ICON-EU behaves similarly. Neither domain nor model class alone separates the outcomes: the negative group consists of global learned systems and ensemble means, and the positive group of regional learned systems and deterministic global numerical models. The experiment cannot separate resolution and domain from training objective, ensemble averaging, model family, or vendor.

Bias correction is part of this conclusion, not a technical footnote. It removes an offset that operational users can estimate from prior valid times, but it also changes several upper-wind-tail rankings and reverses the sign of the AIFS and AIFS ENS temperature comparisons. Reporting raw and corrected values bounds the claim: the global learned penalty appears in nine of ten raw model–variable cells, whereas the regional wind exception is not visible without correction.

The conditional analysis resolves a second ambiguity. Conditioning on the realized observation necessarily creates apparent movement toward the centre, so its sign should not be interpreted as a model defect. The useful information is the cross-model magnitude and the forecast-conditioned reliability view. For upper-tail temperature, the latter is near zero across systems even when observation-conditioned MAE differs substantially.

Finally, ensemble means are point summaries. The similar upper-temperature tail losses of AIFS ENS and ECMWF ENS are consistent with ensemble averaging, not evidence against proper-score training. Probabilistic scores are required to evaluate the objective for which these systems were trained.

6 LIMITATIONS

Some authors have a potential conflict of interest with a subset of the evaluated systems; details are withheld for double-blind review and will be disclosed in the camera-ready. Public documentation states that the EPT-2 family was pretrained partly on SYNOP and METAR networks (Jua.ai, 2026), so station identity may overlap training even though the scored forecast period is later. All reported results are out of sample with respect to model development. Potential vendor effects remain, and no causal attribution is made from the regional counterexamples.

The scored network is geographically uneven: France contributes 39–48% of upper-tail pairs, while several countries have fewer than twenty stations. Monthly jackknife errors do not capture this spatial heterogeneity. IFS, AIFS, and AIFS ENS changed operational versions on 12 May 2026 (European Centre for Medium-Range Weather Forecasts, 2026). The analysis treats each named product as one forecast system and does not isolate version effects.

Results are limited to Europe, ten months, and primarily 6–48 h. The climatological upper-temperature regime contains 13.5% of evaluation hours and is not an impact-defined extreme. Threshold-definition sensitivity is reported because it materially changes magnitudes.

All ensembles are evaluated through their means. The study does not assess calibration, spread, spatial coherence, or event probabilities. The observed-tail MAE comparison is also subject to the forecaster’s dilemma and should not be read as a proper probabilistic evaluation.

7 CONCLUSION

At European stations, 6–48 h leads, and fixed climatological regimes, the tested global learned systems all lose relative MAE skill from all conditions to the upper wind and temperature tails after common domain-mean bias removal. Regional learned and numerical systems provide counterexamples, although the regional wind result depends on the correction and regime definition. The evidence therefore supports a structured contrast in which global learned systems and ensemble means lose tail skill while regional learned and deterministic global numerical systems do not, rather than a uniform learned-versus-numerical divide. Observation-conditioned displacement toward the centre

324 is expected for any imperfect forecast; forecast-conditioned bias shows much smaller model-specific
325 reliability errors. These conclusions concern point summaries and do not extend to probabilistic cal-
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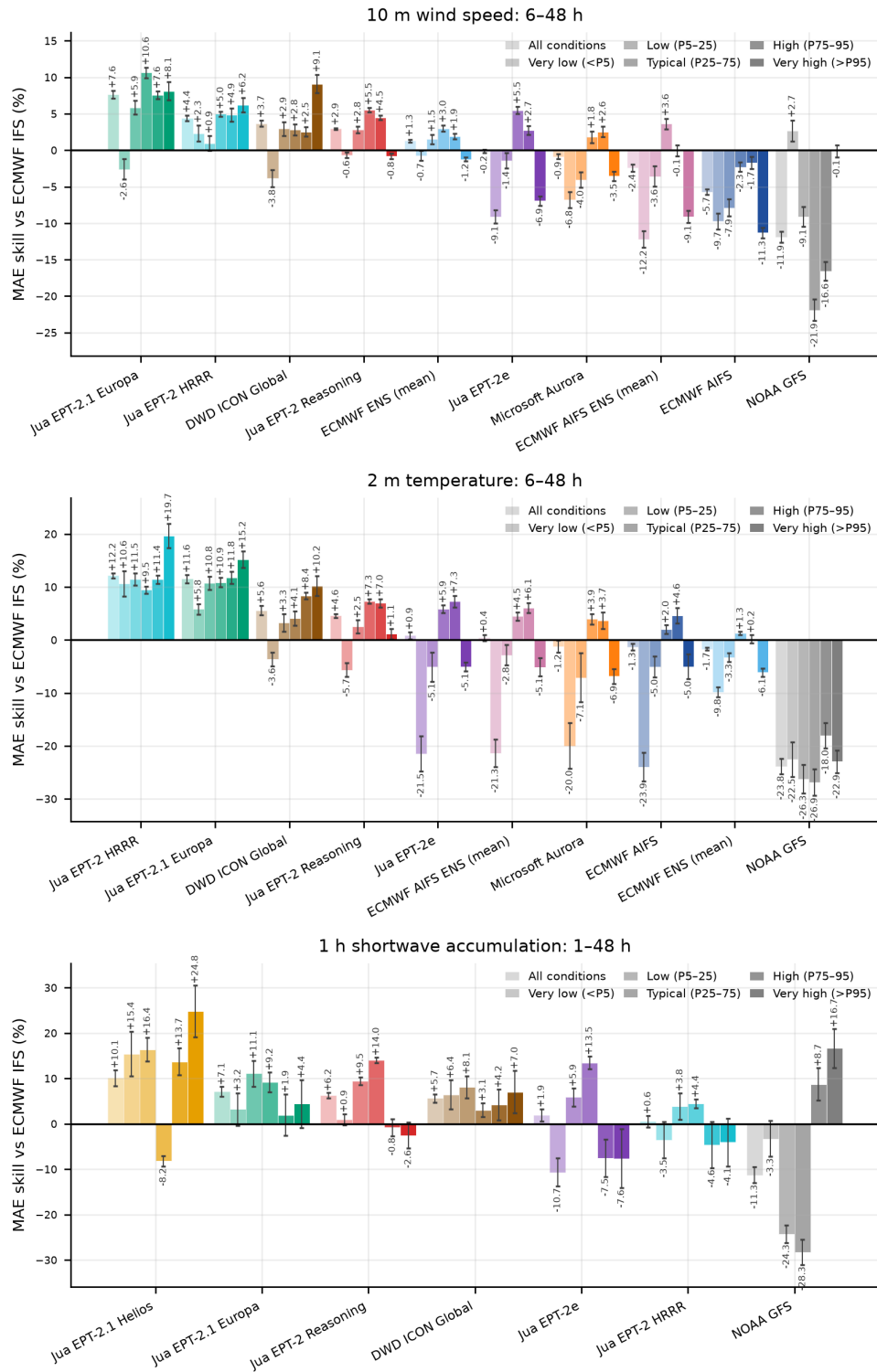


Figure 2: Bias-corrected Track A MAE skill against ECMWF IFS by observation-conditioned climatological regime. Wind and temperature pool 6-48 h; solar pools 1-48 h. Error bars give one leave-one-month-out jackknife standard error.

Conditional bias at 48 h after domain-mean bias correction

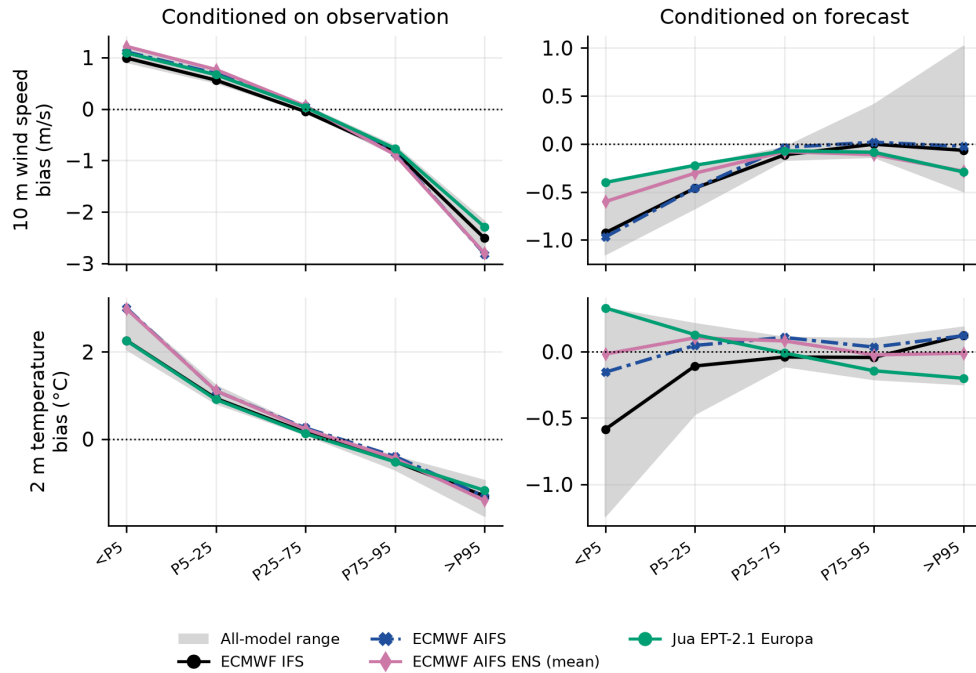


Figure 3: Forecast-minus-observation bias at 48 h after domain-mean bias correction. Left: regimes defined by the observation. Right: regimes defined by the forecast. Grey bands span all models; lines highlight IFS, deterministic AIFS, AIFS ENS, and Europa. The columns use different vertical scales.

REPRODUCIBILITY STATEMENT

The anonymous package reproduces score-based tables and figures from compact aggregate data and includes the extraction and aggregation code. Rebuilding synoptic aggregates from station-level errors requires access to the read-only verification warehouse and is not available to third parties. The package does not release station identities or per-station series. Every preserved compact input is integrity-checked and recorded in the aggregate status file.

AI USE STATEMENT

Generative AI tools were used to review experimental methodology, challenge claims, assist with software changes, inspect generated results, search literature, suggest paper structure, and draft and edit manuscript text. They were not used to generate synthetic observations, formulate or prove mathematical claims, or replace the authors' quantitative analysis. The authors checked AI-assisted code through executable tests and regenerated artifacts, checked numerical claims against aggregate data, reviewed the cited literature and all manuscript text, and take responsibility for the final content.

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A MODEL AND DATA DETAILS

A.1 FORECAST SYSTEMS

Table 1 lists the systems. All reported results are out of sample with respect to model development. IFS, AIFS, and AIFS ENS changed operational versions on 12 May 2026 (European Centre for Medium-Range Weather Forecasts, 2026); the analysis treats each named product as one system and does not isolate version effects.

Table 1: Forecast systems. Ensemble systems are evaluated through their means. ICON-EU appears only in Track B; Helios only in the solar comparison.

Model	Class	Output	Domain
ECMWF IFS (reference)	physical	deterministic	global
ECMWF ENS	physical	ensemble	global
NOAA GFS	physical	deterministic	global
DWD ICON Global	physical	deterministic	global
DWD ICON-EU	physical	deterministic	regional
ECMWF AIFS	regression AI	deterministic	global
ECMWF AIFS ENS	proper-score AI	51-member ensemble	global
Microsoft Aurora	regression AI	deterministic	global
Jua EPT-2 Reasoning	regression AI	deterministic	global
Jua EPT-2e	regression AI	ensemble	global
Jua EPT-2 HRRR	generative AI	ensemble	regional
Jua EPT-2.1 Europa	generative AI	ensemble	regional
Jua EPT-2.1 Helios	solar AI	deterministic	regional

A.2 STATION PANEL AND TRACKS

The scored synoptic panel consists of the 667 stations for which fixed ERA5 thresholds are available. Country counts are in Table 2; the full station catalogue shown in Figure 4 is larger and is not the scored count.

Table 2: Synoptic stations contributing to the fixed-climatology analysis.

Country	Stations with fixed thresholds
AT	8
BE	12
CH	16
CZ	11
DE	67
DK	12
ES	40
FR	227
GB	74
IT	81
NL	13
NO	31
PL	75
Total	667

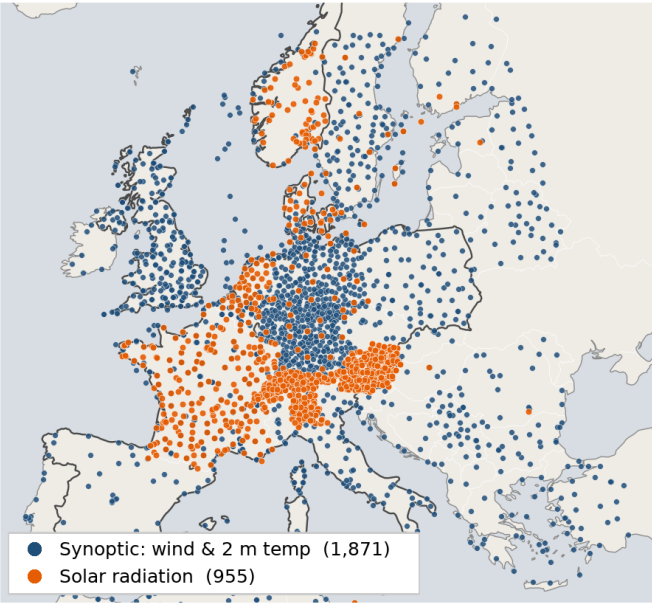


Figure 4: Full station catalogues available to the verification system. Only stations with fixed climatological thresholds enter the reported wind and temperature regimes; the solar catalogue is likewise larger than its scored regime panel.

Track A covers 1 September 2025–30 June 2026 over DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, and DK. It uses all four daily initialization cycles and the common six-hourly lead grid from 6 to 48 h. Track B covers March–June 2026 and compares the regional systems on their common hourly lead grid.

Solar radiation uses hourly leads from 1 to 48 h. Regime-stratified results use 543 stations with fixed thresholds in BE, CH, DE, DK, ES, FR, IT, NL, and NO. Austrian catalogue stations were not part of the frozen score extraction; GB and PL are absent, and CZ lacks fixed solar thresholds. Spain contributes one station through March 2026, and Italian coverage ends after February.

A.3 OBSERVED-VALUE REGIMES

For each station, calendar day, and UTC hour, ERA5 P5, P25, P75, and P95 are estimated over 1991–2020 with a cyclic ± 15 -day window. The five regimes are assigned from the verifying observation.

Because they follow the day-of-year climatology, they represent local anomalies rather than fixed physical severity.

B COMPLETE TRACK A RESULTS

Table 3: Bias-corrected Track A MAE skill (%) against IFS, 6–48 h, by wind regime. Parentheses give leave-one-month-out jackknife standard errors.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.6 (0.5)	-2.6 (1.4)	+5.9 (1.0)	+10.6 (0.7)	+7.6 (0.5)	+8.1 (1.2)
Jua EPT-2 HRRR	+4.4 (0.4)	+2.3 (1.1)	+0.9 (1.0)	+5.0 (0.3)	+4.9 (0.9)	+6.2 (1.0)
Jua EPT-2 Reasoning	+2.9 (0.1)	-0.6 (0.4)	+2.8 (0.5)	+5.5 (0.3)	+4.5 (0.3)	-0.8 (0.4)
Jua EPT-2e	-0.2 (0.2)	-9.1 (0.9)	-1.4 (1.1)	+5.5 (0.5)	+2.7 (0.6)	-6.9 (0.6)
Microsoft Aurora	-0.9 (0.3)	-6.8 (1.1)	-4.0 (1.1)	+1.8 (0.8)	+2.6 (0.7)	-3.5 (0.6)
ECMWF AIFS ENS (mean)	-2.4 (0.5)	-12.2 (1.1)	-3.6 (1.4)	+3.6 (0.7)	-0.1 (0.7)	-9.1 (0.8)
ECMWF AIFS	-5.7 (0.4)	-9.7 (1.1)	-7.9 (1.2)	-2.3 (0.6)	-1.7 (0.8)	-11.3 (0.7)
ECMWF ENS (mean)	+1.3 (0.2)	-0.7 (0.7)	+1.5 (0.6)	+3.0 (0.4)	+1.9 (0.4)	-1.2 (0.2)
NOAA GFS	-11.9 (0.7)	+2.7 (1.5)	-9.1 (1.4)	-21.9 (1.5)	-16.6 (1.3)	-0.1 (0.8)
DWD ICON Global	+3.7 (0.4)	-3.8 (1.2)	+2.9 (0.9)	+2.8 (0.8)	+2.5 (0.6)	+9.1 (1.2)

Table 4: Bias-corrected Track A MAE skill (%) against IFS, 6–48 h, by temperature regime.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+11.6 (0.8)	+5.8 (1.0)	+10.8 (1.2)	+10.9 (0.9)	+11.8 (1.2)	+15.2 (1.5)
Jua EPT-2 HRRR	+12.2 (0.4)	+10.6 (2.4)	+11.5 (1.1)	+9.5 (0.7)	+11.4 (0.8)	+19.7 (2.3)
Jua EPT-2 Reasoning	+4.6 (0.4)	-5.7 (1.3)	+2.5 (1.2)	+7.3 (0.4)	+7.0 (0.8)	+1.1 (1.0)
Jua EPT-2e	+0.9 (0.7)	-21.5 (3.3)	-5.1 (2.8)	+5.9 (0.7)	+7.3 (1.1)	-5.1 (0.9)
Microsoft Aurora	-1.2 (1.1)	-20.0 (4.3)	-7.1 (4.6)	+3.9 (1.0)	+3.7 (1.6)	-6.9 (1.4)
ECMWF AIFS ENS (mean)	+0.4 (0.5)	-21.3 (2.6)	-2.8 (1.9)	+4.5 (0.8)	+6.1 (0.9)	-5.1 (1.7)
ECMWF AIFS	-1.3 (0.6)	-23.9 (2.7)	-5.0 (2.0)	+2.0 (0.8)	+4.6 (1.5)	-5.0 (2.3)
ECMWF ENS (mean)	-1.7 (0.3)	-9.8 (0.9)	-3.3 (0.8)	+1.3 (0.3)	+0.2 (0.8)	-6.1 (0.8)
NOAA GFS	-23.8 (1.4)	-22.5 (3.2)	-26.3 (2.7)	-26.9 (2.5)	-18.0 (2.4)	-22.9 (2.1)
DWD ICON Global	+5.6 (0.9)	-3.6 (1.3)	+3.3 (1.6)	+4.1 (1.3)	+8.4 (0.7)	+10.2 (1.9)

Table 5: Bias-corrected Track A solar MAE skill (%) against IFS, 1–48 h, by regime.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+10.1 (1.7)	+15.4 (4.9)	+16.4 (2.7)	-8.2 (1.2)	+13.7 (3.0)	+24.8 (5.7)
Jua EPT-2.1 Europa	+7.1 (1.1)	+3.2 (3.6)	+11.1 (2.9)	+9.2 (2.2)	+1.9 (4.6)	+4.4 (5.3)
Jua EPT-2 HRRR	+0.6 (1.3)	-3.5 (3.9)	+3.8 (3.0)	+4.4 (1.0)	-4.6 (5.1)	-4.1 (5.2)
Jua EPT-2 Reasoning	+6.2 (0.6)	+0.9 (1.2)	+9.5 (0.8)	+14.0 (0.6)	-0.8 (1.8)	-2.6 (2.8)
Jua EPT-2e	+1.9 (1.4)	-10.7 (3.1)	+5.9 (2.0)	+13.5 (1.4)	-7.5 (4.1)	-7.6 (6.5)
NOAA GFS	-11.3 (1.8)	-3.3 (3.9)	-24.3 (1.9)	-28.3 (2.8)	+8.7 (3.5)	+16.7 (4.3)
DWD ICON Global	+5.7 (0.9)	+6.4 (3.2)	+8.1 (2.4)	+3.1 (1.6)	+4.2 (3.4)	+7.0 (4.7)

Table 6: Track A MAE skill (%) by lead for 10 m wind.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+7.6	+7.8	+8.0	+8.4	+7.7	+8.2	+7.3	+8.0
EPT-2 HRRR	+3.3	+5.0	+3.8	+5.7	+4.5	+6.1	+5.0	+6.9
EPT-2 Reasoning	+1.9	-0.8	+2.3	-0.9	+2.9	-1.0	+3.9	-0.6
EPT-2e	-1.3	-8.4	-1.1	-7.9	-0.3	-7.1	+1.0	-5.9
Aurora	-1.7	-4.2	-1.7	-3.8	-1.1	-3.7	+0.3	-3.0
AIFS ENS	-3.7	-8.6	-3.5	-9.0	-2.8	-9.4	-0.8	-8.9
AIFS	-6.8	-11.5	-6.9	-11.8	-6.0	-11.9	-4.2	-10.3
ENS	+0.2	-1.1	+0.5	-1.1	+1.1	-1.1	+2.3	-1.7
GFS	-11.2	+1.2	-11.3	+1.0	-12.0	-0.1	-12.4	-1.2
ICON Global	+4.6	+9.7	+4.6	+9.7	+4.0	+9.2	+2.5	+8.2

Table 7: Track A MAE skill (%) by lead for 2 m temperature.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.3	+12.7	+12.0	+16.0	+11.5	+14.7	+11.7	+14.6
EPT-2 HRRR	+10.1	+19.6	+12.6	+21.4	+12.1	+19.5	+12.3	+18.5
EPT-2 Reasoning	+2.2	-0.5	+3.5	+0.2	+4.5	+0.9	+6.1	+2.3
EPT-2e	-4.1	-10.0	-1.5	-7.5	+1.2	-4.8	+3.2	-2.8
Aurora	-4.5	-7.7	-2.2	-7.0	-1.1	-7.0	+0.3	-6.7
AIFS ENS	-4.0	-10.2	-1.0	-6.6	+0.3	-4.9	+2.5	-3.3
AIFS	-4.5	-7.6	-2.4	-5.3	-1.6	-5.1	+0.4	-3.8
ENS	-2.9	-6.8	-2.5	-6.3	-1.8	-6.1	-0.5	-5.7
GFS	-27.4	-24.6	-24.3	-22.7	-24.3	-23.3	-22.4	-22.8
ICON Global	+5.3	+10.2	+7.0	+11.7	+6.2	+10.3	+4.1	+8.8

C ROBUSTNESS CHECKS

C.1 RAW AND CORRECTED FORECASTS

Tables 9 and 10 show that the domain-mean correction materially changes wind-tail rankings and reverses the sign of the AIFS temperature-tail comparison. The tail-minus-all column is the direct test used in the main paper.

C.2 THRESHOLD DEFINITION

The variants use the same forecast pairs but different regime constructions. The fixed definition uses station-day-hour anomaly thresholds; the in-window definition uses one country-level percentile distribution over stations and times and therefore selects absolute tails. The large climatological lower-temperature-tail losses of several learned systems collapse under the in-window definition, and several tail-minus-all signs also change.

Table 8: Track A MAE skill (%) by lead for solar radiation.

Model	6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+10.8	+19.1	+9.0	+15.2	+8.1	+15.4	+9.1	+15.5
EPT-2.1 Europa	+5.7	+10.1	+6.8	+9.9	+6.2	+10.2	+6.4	+10.2
ICON Global	+5.7	+9.8	+6.1	+8.7	+5.3	+8.1	+4.3	+6.3
EPT-2 Reasoning	+4.2	+7.0	+5.1	+8.5	+6.0	+9.5	+7.8	+11.1
EPT-2e	-4.5	+2.4	-2.5	+2.1	+0.1	+2.9	+4.1	+8.1
EPT-2 HRRR	-4.8	-0.5	-2.6	+1.2	-1.7	+2.6	+1.0	+5.1
GFS	-13.8	-30.2	-12.5	-29.5	-11.8	-25.8	-11.1	-24.3

Table 9: Track A wind MAE skill (%): raw and bias-corrected all-condition and >P95 values. Tail – all is computed after correction; parentheses give its monthly jackknife standard error.

Model	Raw all	Raw >P95	Corrected all	Corrected >P95	Tail – all
EPT-2.1 Europa	+0.4	-5.8	+7.6	+8.1	+0.5 (0.8)
EPT-2 HRRR	+4.1	+4.0	+4.4	+6.2	+1.8 (0.7)
EPT-2 Reasoning	+2.8	-0.2	+2.9	-0.8	-3.7 (0.5)
EPT-2e	-1.0	-7.0	-0.2	-6.9	-6.8 (0.8)
Aurora	-1.0	-1.9	-0.9	-3.5	-2.7 (0.9)
AIFS ENS	-2.1	-4.9	-2.4	-9.1	-6.7 (0.9)
AIFS	-6.5	-12.3	-5.7	-11.3	-5.6 (1.0)
ENS	+3.2	+4.7	+1.3	-1.2	-2.5 (0.4)
GFS	-8.4	+13.1	-11.9	-0.1	+11.8 (1.3)
ICON Global	+0.2	-0.5	+3.7	+9.1	+5.4 (1.0)

D REGIONAL AND COUNTRY-LEVEL RESULTS

E ADDITIONAL METHODOLOGICAL DETAILS

For each model, valid week, initialization hour, and lead, the additive correction is estimated by pooling station errors from the preceding four valid weeks. It is one Europe-wide scalar, not a station-specific correction. Regime labels depend only on the observation and are unchanged by correction.

If $\hat{S}$ is the full estimate and $\hat{S}_{(-j)}$ omits month j , the reported standard error is

$$SE_{JK} = \sqrt{\frac{M-1}{M} \sum_{j=1}^M \left(\hat{S}_{(-j)} - \bar{\hat{S}}_{(-\cdot)} \right)^2}.$$

MAE is pooled linearly by sample count in both the point estimate and each leave-one-month-out replicate. Track A has $M = 10$ and Track B $M = 4$. These values describe sensitivity to months; they are not exact confidence intervals and do not capture country-level dependence.

Table 10: As Table 9, for temperature.

Model	Raw all	Raw >P95	Corrected all	Corrected >P95	Tail – all
EPT-2.1 Europa	+12.3	+18.1	+11.6	+15.2	+3.6 (1.2)
EPT-2 HRRR	+8.1	+15.8	+12.2	+19.7	+7.5 (2.1)
EPT-2 Reasoning	+3.9	-0.2	+4.6	+1.1	-3.5 (0.9)
EPT-2e	+1.1	-5.4	+0.9	-5.1	-5.9 (1.2)
Aurora	-3.4	-9.7	-1.2	-6.9	-5.6 (1.8)
AIFS ENS	+4.7	+2.9	+0.4	-5.1	-5.5 (1.7)
AIFS	+3.4	+4.4	-1.3	-5.0	-3.7 (2.0)
ENS	-4.4	-11.4	-1.7	-6.1	-4.4 (0.6)
GFS	-28.1	-29.0	-23.8	-22.9	+0.9 (2.3)
ICON Global	+9.1	+16.9	+5.6	+10.2	+4.7 (1.7)

Table 11: Bias-corrected tail skill (%) under fixed 1991–2020 climatological thresholds (Clim.) and percentiles of the evaluation window (Window).

Model	Wind >P95		Temperature >P95		Temperature <P5	
	Clim.	Window	Clim.	Window	Clim.	Window
EPT-2.1 Europa	+8.1	+4.9	+15.2	+14.3	+5.8	+10.6
EPT-2 HRRR	+6.2	+2.2	+19.7	+12.7	+10.6	+11.7
EPT-2 Reasoning	-0.8	+0.4	+1.1	+6.9	-5.7	+4.3
EPT-2e	-6.9	-3.8	-5.1	-0.5	-21.5	+0.3
Aurora	-3.5	-1.0	-6.9	-0.5	-20.0	-1.5
AIFS ENS	-9.1	-6.9	-5.1	-6.3	-21.3	-0.3
AIFS	-11.3	-6.1	-5.0	-11.7	-23.9	-2.0
ENS	-1.2	-0.3	-6.1	-3.3	-9.8	-2.0
GFS	-0.1	+3.6	-22.9	-31.1	-22.5	-25.8
ICON Global	+9.1	+5.9	+10.2	+10.1	-3.6	+3.9

Table 12: Track B wind MAE skill (%), 1–48 h, by regime.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+6.5 (0.3)	-0.7 (0.9)	+2.4 (0.3)	+8.3 (0.5)	+7.3 (0.3)	+9.3 (1.5)
Jua EPT-2 HRRR	+2.5 (0.7)	+4.4 (1.1)	-2.4 (1.7)	+3.4 (0.5)	+1.9 (0.2)	+4.2 (1.7)
DWD ICON-EU	+4.6 (0.4)	-1.5 (1.1)	+1.2 (0.6)	+5.0 (0.5)	+3.1 (0.2)	+10.4 (1.8)

Table 13: Track B temperature MAE skill (%), 1–48 h, by regime.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+13.4 (0.6)	+7.9 (0.9)	+11.9 (1.5)	+12.5 (1.3)	+13.5 (0.4)	+16.8 (2.0)
Jua EPT-2 HRRR	+12.2 (0.6)	+14.6 (2.3)	+15.0 (1.2)	+9.2 (0.3)	+8.8 (1.0)	+17.5 (3.3)
DWD ICON-EU	+12.5 (0.8)	+9.5 (0.7)	+11.9 (1.4)	+10.7 (1.2)	+11.3 (1.4)	+17.4 (2.8)

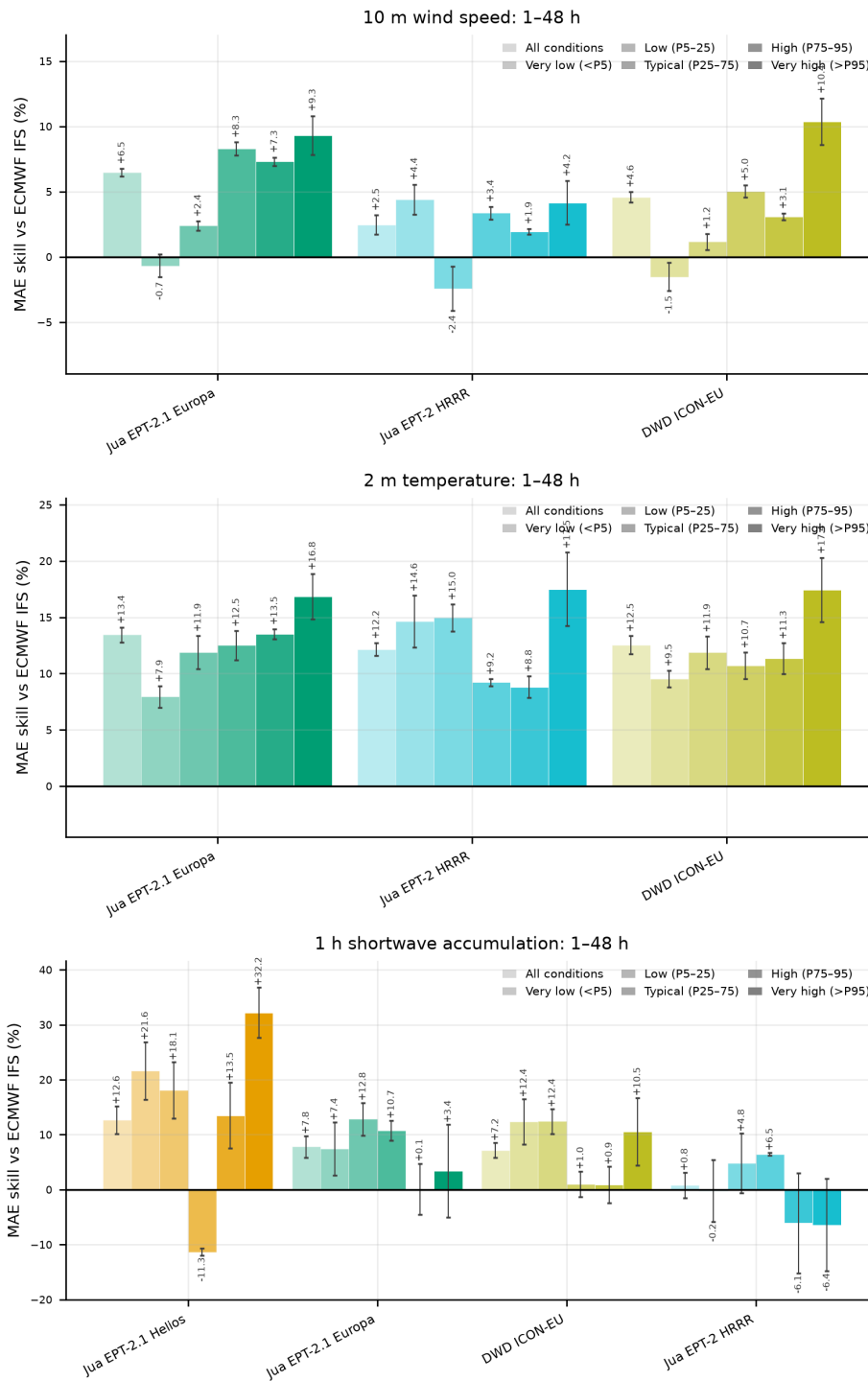


Figure 5: Track B bias-corrected MAE skill against IFS by observation-conditioned regime, March–June 2026, 1–48 h. Uncertainty uses four monthly blocks.

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Table 14: Track B solar MAE skill (%), 1–48 h, by regime.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+12.6 (2.5)	+21.6 (5.2)	+18.1 (5.1)	-11.3 (0.6)	+13.5 (5.9)	+32.2 (4.6)
Jua EPT-2.1 Europa	+7.8 (2.0)	+7.4 (4.8)	+12.8 (3.0)	+10.7 (1.8)	+0.1 (4.7)	+3.4 (8.4)
Jua EPT-2 HRRR	+0.8 (2.3)	-0.2 (5.6)	+4.8 (5.4)	+6.5 (0.3)	-6.1 (9.1)	-6.4 (8.4)
DWD ICON-EU	+7.2 (1.4)	+12.4 (4.1)	+12.4 (2.2)	+1.0 (2.3)	+0.9 (3.3)	+10.5 (6.2)